

Statistics 652: Fall - 2026

**A BRIEF INTRODUCTION TO
ROBUST STATISTICS**

David E. Tyler

Rutgers

The State University of New Jersey

References

- Huber, P.J. (1981). *Robust Statistics*. Wiley, New York.
- Hampel, F.R., Ronchetti, E.M., Rousseeuw, P.J. and Stahel, W.A. (1986). *Robust Statistics: The Approach Based on Influence Functions*. Wiley, New York.
- Maronna, R.A., Martin, R.D. and Yohai, V.J. (2006). *Robust Statistics: Theory and Methods*. Wiley, New York.

PART 1
CONCEPTS AND BASIC METHODS

MOTIVATION

- Data Set: $X_1, X_2, \dots, X_n$
- Parametric Model: $F(x_1, \dots, x_n \mid \theta)$

θ : Unknown parameter

F : Known function

- e.g. $X_1, X_2, \dots, X_n$ i.i.d. $Normal(\mu, \sigma^2)$

Q: Is it realistic to believe we don't know (μ, σ^2) , but we know e.g. the shape of the tails of the distribution?

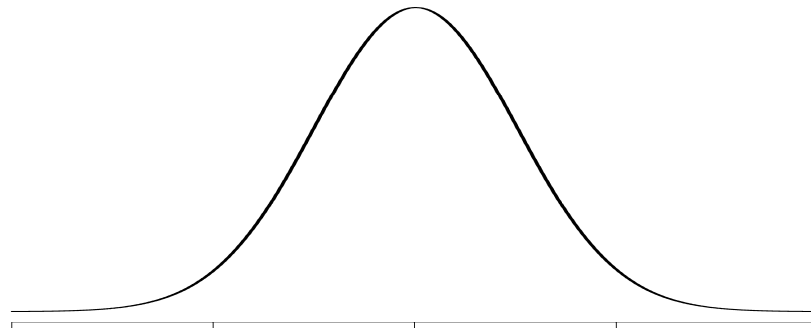
A: The model is assumed to be approximately true, e.g. symmetric and unimodal (past experience).

Q: Are statistical methods which are good under the model reasonably good if the model is only approximately true?

ROBUST STATISTICS

Formally addresses this issue.

CLASSIC EXAMPLE: MEAN .vs. MEDIAN

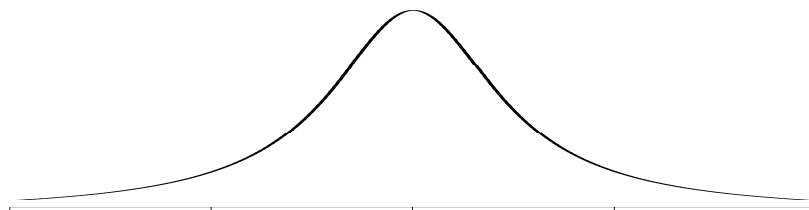


- Symmetric distributions: $\mu = \begin{cases} \text{population mean} \\ \text{population median} \end{cases}$
- Sample mean: $\bar{X} \approx \text{Normal} \left(\mu, \frac{\sigma^2}{n} \right)$
- Sample median: $\text{Median} \approx \text{Normal} \left(\mu, \frac{1}{n} \frac{1}{4f(\mu)^2} \right)$
- At normal: $\text{Median} \approx \text{Normal} \left(\mu, \frac{\sigma^2}{n} \frac{\pi}{2} \right)$
- Asymptotic Relative Efficiency of Median to Mean

$$ARE(\text{Median}, \bar{X}) = \frac{\text{avar}(\bar{X})}{\text{avar}(\text{Median})} = \frac{2}{\pi} = 0.6366$$

CAUCHY DISTRIBUTION

$$X \sim \text{Cauchy}(\mu, \sigma^2)$$



$$f(x; \mu, \sigma) = \frac{1}{\pi\sigma} \left(1 + \left(\frac{x - \mu}{\sigma} \right)^2 \right)^{-1}$$

- Mean: $\bar{X} \sim \text{Cauchy}(\mu, \sigma^2)$ • Median $\approx \text{Normal} \left(\mu, \frac{\pi^2 \sigma^2}{4n} \right)$
- $ARE(\text{Median}, \bar{X}) = \infty$ or $ARE(\bar{X}, \text{Median}) = 0$
- For t on ν degrees of freedom:

$$ARE(\text{Median}, \bar{X}) = \frac{4}{(\nu - 2)\pi} \frac{\Gamma((\nu + 1)/2)^2}{\Gamma(\nu/2)}$$

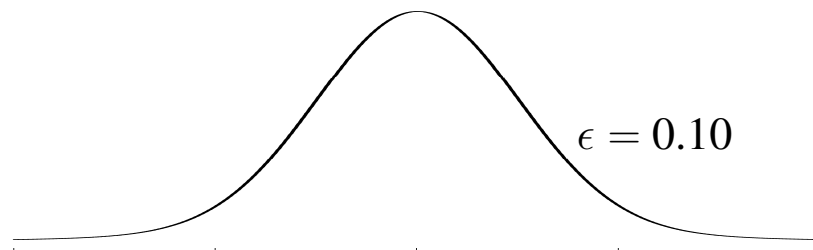
	$\nu \leq 2$	3	4	5
$ARE(\text{Median}, \bar{X})$	∞	1.621	1.125	0.960
$ARE(\bar{X}, \text{Median})$	0	0.617	0.888	1.041

MIXTURE OF NORMALS

Theory of errors: Central Limit Theorem gives plausibility to normality.

$$X \sim \begin{cases} \text{Normal}(\mu, \sigma^2) & \text{with probability } 1 - \epsilon \\ \text{Normal}(\mu, (3\sigma)^2) & \text{with probability } \epsilon \end{cases}$$

i.e. not all measurements are equally precise.



$$X \sim (1 - \epsilon) \text{Normal}(\mu, \sigma^2) + \epsilon \text{Normal}(\mu, (3\sigma)^2)$$

- Classic paper: Tukey (1960), A survey of sampling from contaminated distributions.
 - For $\epsilon > 0.10 \Rightarrow ARE(\text{Median}, \bar{X}) > 1$
 - The mean absolute deviation is more efficient than the sample standard deviation for $\epsilon > 0.01$.

PRINCETON ROBUSTNESS STUDIES

Andrews, et.al. (1972)

Other estimates of location.

- α -trimmed mean: Trim a proportion of α from both ends of the data set and then take the mean. (Throwing away data?)
- α -Windsorized mean: Replace a proportion of α from both ends of the data set by the next closest observation and then take the mean.
- Example: 2, 4, 5, 10, 200
Mean = 44.2 Median = 5
20% trimmed mean = $(4 + 5 + 10) / 3 = 6.33$
20% Windsorized mean = $(4 + 4 + 5 + 10 + 10) / 5 = 6.6$

Measuring the robustness of a statistic

- **Relative Efficiency over a range of distributional models.**
 - There exist estimates of location which are asymptotically most efficient for the center of any symmetric distribution. (Adaptive estimation, semi-parametrics).
 - **Resistant to contamination?**

Measuring the resistance/robustness of a statistic

- **Influence Function over a range of distributional models.**
- **Maximum Bias Function and the Breakdown Point.**

Measuring the effect of an outlier (*not modeled*)

- *Good Data Set:* $x_1, \dots, x_{n-1}$

$$\text{Statistic: } T_{n-1} = T(x_1, \dots, x_{n-1})$$

- *Contaminated Data Set:* $x_1, \dots, x_{n-1}, \mathbf{x}$

$$\text{Contaminated Value: } T_n = T(x_1, \dots, x_{n-1}, \mathbf{x})$$

THE SENSITIVITY CURVE (*Tukey, 1970*)

$$SC_n(\mathbf{x}) = n(T_n - T_{n-1}) \Leftrightarrow T_n = T_{n-1} + \frac{1}{n}SC_n(\mathbf{x})$$

THE INFLUENCE FUNCTION (*Hampel, 1969, 1974*)

Population version of the sensitivity curve.

THE INFLUENCE FUNCTION

- Statistic $T_n = T(F_n)$ estimates $T(F)$.
- Consider F and the ϵ -contaminated distribution,

$$F_\epsilon = (1 - \epsilon)F + \epsilon\delta_{\mathbf{x}}$$

where $\delta_{\mathbf{x}}$ is the point mass distribution at $\mathbf{x}$.



- Compare Functional Values: $T(F)$.vs. $T(F_\epsilon)$
- Given **Qualitative Robustness** (*Continuity*):

$$T(F_\epsilon) \rightarrow T(F) \text{ as } \epsilon \rightarrow 0$$

(e.g. the mode is not qualitatively robust)

- **Influence Function** (*Infinitesimal perturbation: Gâteaux Derivative*)

$$IF(\mathbf{x}; T, F) = \lim_{\epsilon \rightarrow 0} \frac{T(F_\epsilon) - T(F)}{\epsilon} = \left. \frac{\partial}{\partial \epsilon} T(F_\epsilon) \right|_{\epsilon=0}$$

EXAMPLES

- **Mean:** $T(F) = E_F[X]$.

$$\begin{aligned}T(F_\epsilon) &= E_{F_\epsilon}(X) \\&= (1 - \epsilon)E_F[X] + \epsilon E[\delta_{\mathbf{x}}] \\&= (1 - \epsilon)T[F] + \epsilon \mathbf{x}\end{aligned}$$

$$\mathbf{IF}(\mathbf{x}; \mathbf{T}, \mathbf{F}) = \lim_{\epsilon \rightarrow 0} \frac{(1 - \epsilon)T(F) + \epsilon \mathbf{x} - T(F)}{\epsilon} = \mathbf{x} - \mathbf{T}(\mathbf{F})$$

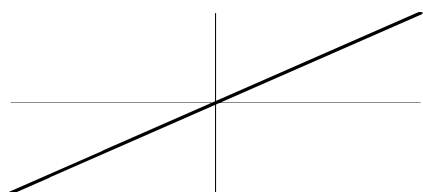
- **Median:** $T(F) = F^{-1}(1/2)$

$$\mathbf{IF}(\mathbf{x}; \mathbf{T}, \mathbf{F}) = \{2 \mathbf{f}(\mathbf{T}(\mathbf{F}))\}^{-1} \mathbf{sign}(\mathbf{X} - \mathbf{T}(\mathbf{F}))$$

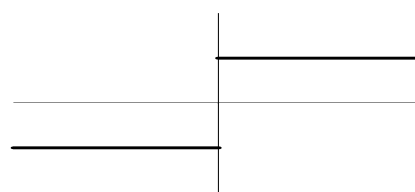
Plots of Influence Functions

Gives insight into the behavior of a statistic.

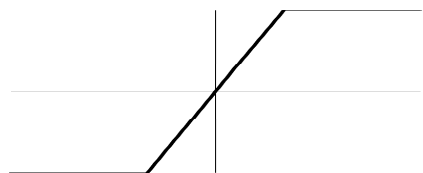
$$T_n \approx T_{n-1} + \frac{1}{n}IF(\mathbf{x}; T, F)$$



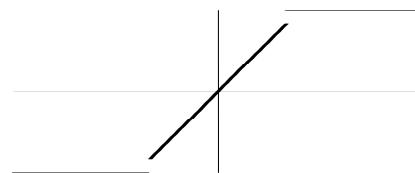
Mean



Median



α -trimmed mean



α -Winsorized mean
(*somewhat unexpected?*)

Desirable properties for the influence function

- **SMALL**

- Gross Error Sensitivity

$$GES(T; F) = \sup_{\mathbf{x}} | IF(\mathbf{x}; T, F) |$$

$$GES < \infty \Rightarrow \text{B-robust (Bias-robust)}$$

- Asymptotic Variance

$$\text{Note : } E_F[IF(X; T, F)] = 0$$

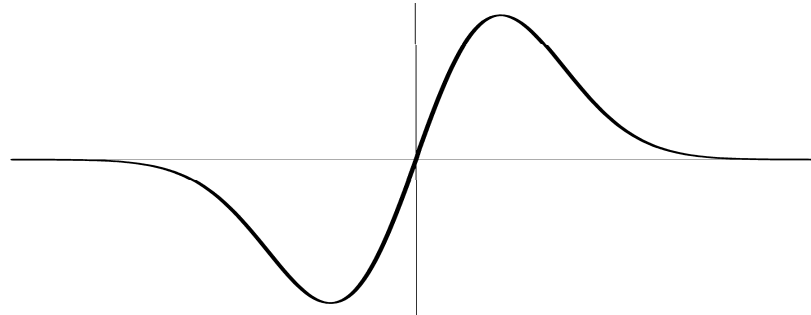
$$AV(T; F) = E_F[IF(X; T, F)^2]$$

- Under general conditions, e.g. *Frèchet differentiability*,

$$\sqrt{n}(T(X_1, \dots, X_n) - T(F)) \rightarrow \text{Normal}(0, AV(T; F))$$

- Trade-off at the normal model: **Smaller AV $\leftrightarrow$ Larger GES**
- **SMOOTH** (local shift sensitivity): protects e.g. against rounding error.
- **REDESCENDING** to 0.

REDESCENDING INFLUENCE FUNCTION



- Example: Data Set of Male Heights in cm

180, 175, 192, ..., 185, 2020, 190, ...

- **Redescender = Automatic Outlier Detector**

CLASSES OF ESTIMATES

L-statistics: Linear combination of order statistics

- Let $X_{(1)} \geq \dots \geq X_{(n)}$ represent the order statistics.

$$T(X_1, \dots, X_n) = \sum_{i=1}^n a_{i,n} X(i)$$

where $a_{i,n}$ are constants.

- **Examples:**
 - **Mean.** $a_{i,n} = 1/n$

$$a_{i,n} = \begin{cases} 1 & i = \frac{n+1}{2} \\ 0 & i \neq \frac{n+1}{2} \end{cases} \quad a_{i,n} = \begin{cases} \frac{1}{2} & i = \frac{n}{2}, \frac{n}{2} + 1 \\ 0 & \text{otherwise} \end{cases}$$

for n odd for n even

- **α -trimmed mean.**

- **α -Winsorized mean.**

- General form for the influence function exists
- Can obtain any desirable monotonic shape, but not redescending
- Do not readily generalize to other settings.

M-ESTIMATES

Huber (1964, 1967)

Maximum likelihood type estimates
under non-standard conditions

• **One-Parameter Case.** $X_1, \dots, X_n$ i.i.d. $f(x; \theta)$, $\theta \in \Theta$

• **Maximum likelihood estimates**

– Likelihood function. $L(\theta \mid x_1, \dots, x_n) = \prod_{i=1}^n f(x_i; \theta)$

– Minimize the negative log-likelihood.

$$\min_{\theta} \sum_{i=1}^n \rho(x_i; \theta) \quad \text{where} \quad \rho(x_i; \theta) = -\log f(x_i; \theta).$$

– Solve the likelihood equations

$$\sum_{i=1}^n \psi(x_i; \theta) = 0 \quad \text{where} \quad \psi(x_i; \theta) = \frac{\partial \rho(x_i; \theta)}{\partial \theta}$$

DEFINITIONS OF M-ESTIMATES

- Objective function approach: $\hat{\theta} = \arg \min \sum_{i=1}^n \rho(x_i; \theta)$
- M-estimating equation approach: $\sum_{i=1}^n \psi(x_i; \hat{\theta}) = 0$.
 - *Note:* Unique solution when $\psi(x; \theta)$ is strictly monotone in θ .

- **Basic examples.**

- **Mean.** MLE for Normal: $f(x) = (2\pi)^{-1/2} e^{-\frac{1}{2}(x-\theta)^2}$ for $x \in \mathbb{R}$.

$$\rho(x; \theta) = (x - \theta)^2 \quad \text{or} \quad \psi(x; \theta) = x - \theta$$

- **Median.** MLE for Double Exponential: $f(x) = \frac{1}{2} e^{-|x-\theta|}$ for $x \in \mathbb{R}$.

$$\rho(x; \theta) = |x - \theta| \quad \text{or} \quad \psi(x; \theta) = \text{sign}(x - \theta)$$

- ρ and ψ need not be related to any density or to each other.
- Estimates can be evaluated under various distributions.

M-ESTIMATES OF LOCATION

A symmetric and translation equivariant M-estimate.

Translation equivariance

$$X_i \rightarrow X_i + a \Rightarrow T_n \rightarrow T_n + a$$

gives

$$\rho(x; t) = \rho(x - t) \text{ and } \psi(x; t) = \psi(x - t)$$

Symmetric

$$X_i \rightarrow -X_i \Rightarrow T_n \rightarrow -T_n$$

gives

$$\rho(-r) = \rho(r) \text{ or } \psi(-r) = -\psi(r).$$

Alternative derivation

Generalization of MLE for center of symmetry for a given family of symmetric distributions.

$$f(x; \theta) = g(|x - \theta|)$$

INFLUENCE FUNCTION OF M-ESTIMATES

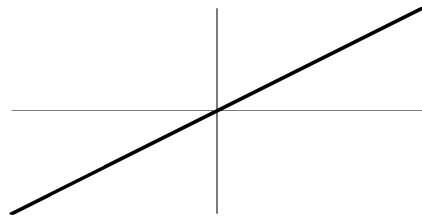
M-FUNCTIONAL: $T(F)$ is the solution to $E_F[\psi(\mathbf{X}; T(F))] = 0$.

$$IF(\mathbf{x}; T, F) = c(T, F)\psi(\mathbf{x}; T(F))$$

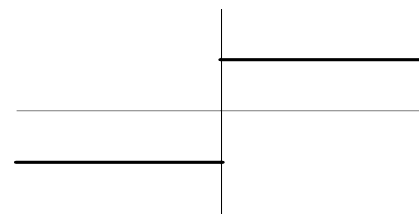
where $c(t, f) = -1/E_F\left[\frac{\partial\psi(\mathbf{X};\theta)}{\partial\theta}\right]$ evaluated at $\theta = T(F)$.

Note: $E_F[IF(X; T, F)] = 0$.

One can decide what shape is desired for the Influence Function and then construct an appropriate M-estimate.

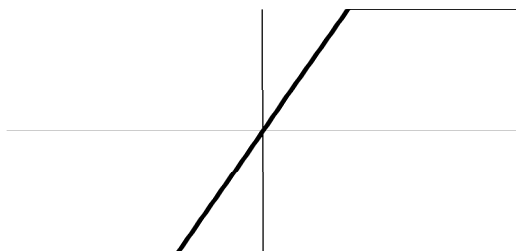


$\Rightarrow$ Mean



$\Rightarrow$ Median

EXAMPLE



Choose

$$\psi(r) = \begin{cases} c & r \geq c \\ r & |r| < c \\ -c & r \leq -c \end{cases}$$

where c is a tuning constant.

Huber's M-estimate
Adaptively trimmed mean.

i.e. *the proportion trimmed depends upon the data.*

DERIVATION OF INFLUENCE FUNCTION FROM M-ESTIMATES

Sketch

Let $T_\epsilon = T(F_\epsilon)$, and so

$$0 = E_{F_\epsilon}[\psi(\mathbf{X}; T_\epsilon)] = (1 - \epsilon)E_F[\psi(\mathbf{X}; T_\epsilon)] + \epsilon \psi(\mathbf{x}; T_\epsilon).$$

Taking the derivative with respect to $\epsilon \Rightarrow$

$$0 = -E_F[\psi(\mathbf{X}; T_\epsilon)] + (1 - \epsilon)\frac{\partial}{\partial \epsilon}E_F[\psi(\mathbf{X}; T_\epsilon)] + \psi(\mathbf{x}; T_\epsilon) + \epsilon\frac{\partial}{\partial \epsilon}\psi(\mathbf{x}; T_\epsilon).$$

Let $\psi'(x, \theta) = \partial\psi(x, \theta)/\partial\theta$. Using the chain rule $\Rightarrow$

$$0 = -E_F[\psi(\mathbf{X}; T_\epsilon)] + (1 - \epsilon)E_F[\psi'(\mathbf{X}; T_\epsilon)]\frac{\partial T_\epsilon}{\partial \epsilon} + \psi(\mathbf{x}; T_\epsilon) + \epsilon\psi'(\mathbf{x}; T_\epsilon)\frac{\partial T_\epsilon}{\partial \epsilon}.$$

Letting $\epsilon \rightarrow 0$ and using qualitative robustness, i.e. $T(F_\epsilon) \rightarrow T(F)$, then gives

$$0 = E_F[\psi'(\mathbf{X}; T)]IF(\mathbf{x}; T(F)) + \psi(\mathbf{x}; T(F)) \Rightarrow \text{RESULTS.}$$

ASYMPTOTIC NORMALITY OF M-ESTIMATES

Sketch

Let $\theta = T(F)$. Using Taylor series expansion on $\psi(x; \hat{\theta})$ about θ gives

$$0 = \sum_{i=1}^n \psi(x_i; \hat{\theta}) = \sum_{i=1}^n \psi(x_i; \theta) + (\hat{\theta} - \theta) \sum_{i=1}^n \psi'(x_i; \theta) + \dots,$$

or

$$0 = \sqrt{n} \left\{ \frac{1}{n} \sum_{i=1}^n \psi(x_i; \theta) \right\} + \sqrt{n}(\hat{\theta} - \theta) \left\{ \frac{1}{n} \sum_{i=1}^n \psi'(x_i; \theta) \right\} + O_p(1/\sqrt{n}).$$

By the CLT, $\sqrt{n} \left\{ \frac{1}{n} \sum_{i=1}^n \psi(x_i; \theta) \right\} \rightarrow_d \mathbf{Z} \sim \text{Normal}(0, E_F[\psi(\mathbf{X}; \theta)^2])$.

By the WLLN, $\frac{1}{n} \sum_{i=1}^n \psi'(x_i; \theta) \rightarrow_p E_F[\psi'(\mathbf{X}; \theta)]$.

Thus, by Slutsky's theorem,

$$\sqrt{n}(\hat{\theta} - \theta) \rightarrow_d -\mathbf{Z} / E_F[\psi'(\mathbf{X}; \theta)] \sim \text{Normal}(0, \sigma^2),$$

where

$$\sigma^2 = E_F[\psi(\mathbf{X}; \theta)^2] / E_F[\psi'(\mathbf{X}; \theta)]^2 = E_F[IF(\mathbf{X}; TF)^2]$$

- **NOTE: Proving Frèchet differentiability is not necessary.**

M-estimates of location

Adaptively weighted means

- Recall translation equivariance and symmetry implies

$$\psi(x; t) = \psi(x - t) \quad \text{and} \quad \psi(-r) = -\psi(r).$$

- Express $\psi(r) = ru(r)$ and let $w_i = u(x_i - \theta)$, then

$$0 = \sum_{i=1}^n \psi(x_i - \theta) = \sum_{i=1}^n (x_i - \theta)u(x_i - \theta) = \sum_{i=1}^n w_i \{x_i - \theta\}$$

$$\Rightarrow \quad \hat{\theta} = \frac{\sum_{i=1}^n \mathbf{w}_i \mathbf{x}_i}{\sum_{i=1}^n \mathbf{w}_i}$$

The weights are determined by the data cloud.

•••••

$\hat{\theta}$

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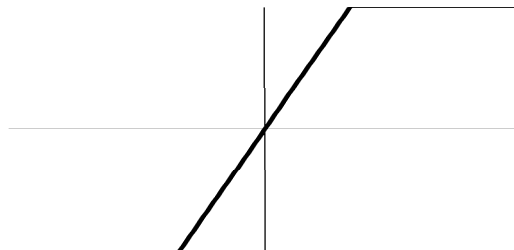
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⏟
Heavily Downweighted

SOME COMMON M-ESTIMATES OF LOCATION

Huber's M-estimate

$$\psi(r) = \begin{cases} c & r \geq c \\ r & |r| < c \\ -c & r \leq -c \end{cases}$$

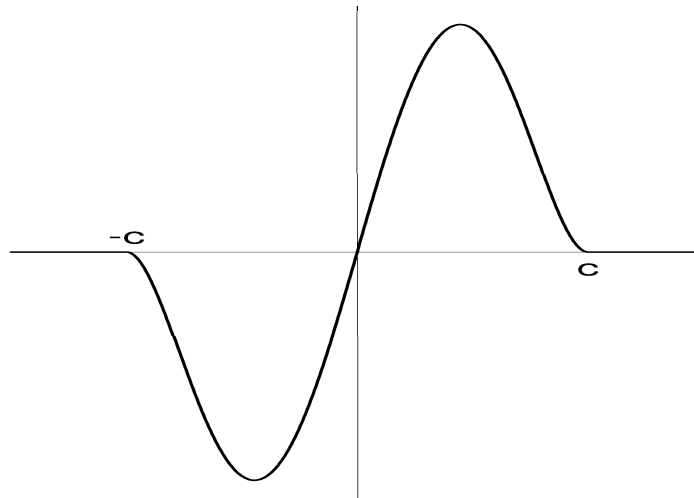


- Given bound on the GES, it has maximum efficiency at the normal model.
- MLE for the “*least favorable distribution*”, i.e. symmetric unimodal model with smallest Fisher Information within a “neighborhood” of the normal.

LFD = Normal in the middle and double exponential in the tails.

Tukey's Bi-weight M-estimate (or bi-square)

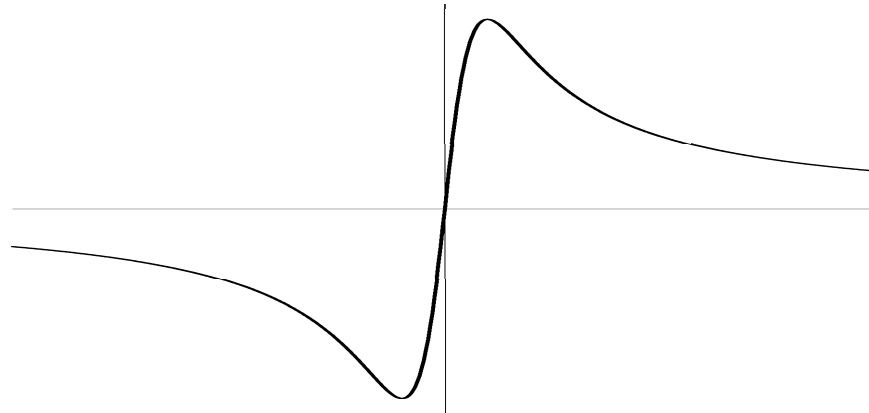
$$u(r) = \left\{ \left(1 - \frac{r^2}{c^2} \right)_+ \right\}^2 \quad \text{where } a_+ = \max\{0, a\}$$



- Linear near zero
- Smooth (continuous second derivatives)
- Strongly redescending to 0
- ***NOT AN MLE.***

CAUCHY MLE

$$\psi(r) = \frac{r/c}{(1 + r^2/c^2)}$$



NOT A STRONG REDESCENDER.

COMPUTATIONS

- **IRLS:** *Iterative Re-weighted Least Squares Algorithm.*

$$\hat{\theta}_{k+1} = \frac{\sum_{i=1}^n w_{i,k} x_i}{\sum_{i=1}^n w_{i,k}}$$

where $w_{i,k} = u(x_i - \hat{\theta}_k)$ and $\hat{\theta}_o$ is any initial value.

- **Re-weighted mean = One-step M-estimate**

$$\hat{\theta}_1 = \frac{\sum_{i=1}^n w_{i,o} x_i}{\sum_{i=1}^n w_{i,o}},$$

where $\hat{\theta}_o$ is a preliminary robust estimate of location, such as the median.

PROOF OF CONVERGENCE

Sketch

$$\hat{\theta}_{k+1} - \hat{\theta}_k = \frac{\sum_{i=1}^n w_{i,k} x_i}{\sum_{i=1}^n w_{i,k}} - \hat{\theta}_k = \frac{\sum_{i=1}^n w_{i,k} (x_i - \hat{\theta}_k)}{\sum_{i=1}^n w_{i,k}} = \frac{\sum_{i=1}^n \psi(x_i - \hat{\theta}_k)}{\sum_{i=1}^n u(x_i - \hat{\theta}_k)}$$

Note: If $\hat{\theta}_k > \hat{\theta}$, then $\hat{\theta}_k > \hat{\theta}_{k+1}$ and $\hat{\theta}_k < \hat{\theta}$, then $\hat{\theta}_k < \hat{\theta}_{k+1}$.

Decreasing objective function

Let $\rho(r) = \rho_o(r^2)$ and suppose $\rho'_o(s) \geq 0$ and $\rho''_o(s) \leq 0$. Then

$$\sum_{i=1}^n \rho(x_i - \hat{\theta}_{k+1}) < \sum_{i=1}^n \rho(x_i - \hat{\theta}_k).$$

- **Examples:**

- Mean $\Rightarrow \rho_o(s) = s$
- Median $\Rightarrow \rho_o(s) = \sqrt{s}$
- Cauchy MLE $\Rightarrow \rho_o(s) = \log(1 + s)$.

- Include redescending M-estimates of location.
- Generalization of EM algorithm for mixture of normals.

PROOF OF MONOTONE CONVERGENCE

Sketch

Let $r_{i,k} = (x_i - \hat{\theta}_k)$. By Taylor series with remainder term,

$$\rho_o(r_{i,k+1}^2) = \rho_o(r_{i,k}^2) + (r_{i,k+1}^2 - r_{i,k}^2)\rho'_o(r_{i,k}^2) + \frac{1}{2}(r_{i,k+1}^2 - r_{i,k}^2)^2\rho''_o(r_{i,*}^2)$$

So,

$$\sum_{i=1}^n \rho_o(r_{i,k+1}^2) < \sum_{i=1}^n \rho_o(r_{i,k}^2) + \sum_{i=1}^n (r_{i,k+1}^2 - r_{i,k}^2)\rho'_o(r_{i,k}^2)$$

Now,

$$ru(r) = \psi(r) = \rho'(r) = 2r\rho'_o(r^2) \quad \text{and so} \quad \rho'_o(r^2) = u(r)/2.$$

Also,

$$(r_{i,k+1}^2 - r_{i,k}^2) = (\hat{\theta}_{k+1} - \hat{\theta}_k)^2 - 2(\hat{\theta}_{k+1} - \hat{\theta}_k)(x_i - \hat{\theta}_k).$$

Thus,

$$\sum_{i=1}^n \rho_o(r_{i,k+1}^2) < \sum_{i=1}^n \rho_o(r_{i,k}^2) - \frac{1}{2}(\hat{\theta}_{k+1} - \hat{\theta}_k)^2 \sum_{i=1}^n u(r_{i,k}) < \sum_{i=1}^n \rho_o(r_{i,k}^2)$$

- **Slow but Sure** $\Rightarrow$ Switch to Newton-Raphson after a few iterations.

PART 2

MORE ADVANCED CONCEPTS AND METHODS

SCALE EQUIVARIANCE

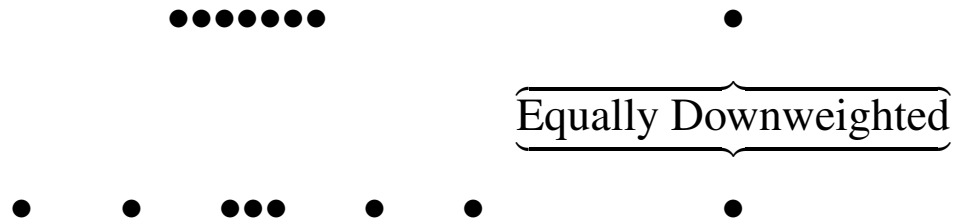
- M-estimates of location alone are not scale equivariant in general, i.e.

$$X_i \rightarrow X_i + a \Rightarrow \hat{\theta} \rightarrow \hat{\theta} + a \quad \text{location equivariant}$$

$$X_i \rightarrow b X_i \not\Rightarrow \hat{\theta} \rightarrow b \hat{\theta} \quad \text{not scale equivariant}$$

(Exceptions: the mean and median.)

- Thus, the adaptive weights are not dependent on the spread of the data



SCALE STATISTICS: s_n

$$X_i \rightarrow bX_i + a \Rightarrow s_n \rightarrow |b| s_n$$

- Sample standard deviation.

$$s_n = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- MAD (or more appropriately MADAM).

Median Absolute Deviation About the Median

$$s_n^* = \text{Median} | x_i - \text{median} |$$

$$s_n = 1.4826 s_n^* \rightarrow_p \sigma \text{ at } \text{Normal}(\mu, \sigma^2)$$

Example

2, 4, 5, 10, 12, 14, 200

- Median = 10
- Absolute Deviations: 8, 6, 5, 0, 2, 4, 190
- MADAM = 5 $\Rightarrow s_n = 7.413$
- (*Standard deviation* = 72.7661)

M-estimates of location with auxiliary scale

$$\widehat{\theta} = \arg \min \sum_{i=1}^n \rho \left(\frac{x_i - \theta}{c s_n} \right)$$

or

$$\widehat{\theta} \text{ solves: } \sum_{i=1}^n \psi \left(\frac{x_i - \theta}{c s_n} \right) = 0$$

- **c: tuning constant**
- **s_n : consistent for σ at normal**
- **Note: simultaneous estimate of scale can be problematic.**

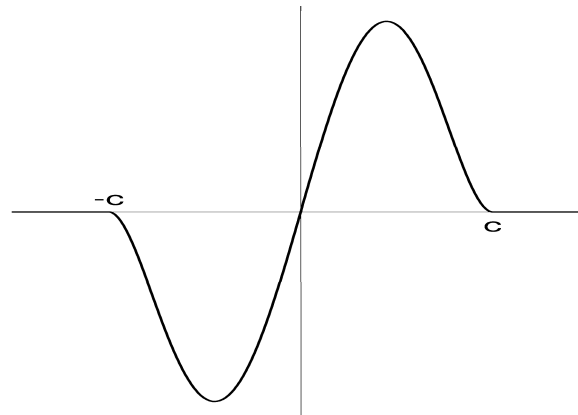
Tuning an M-estimate

- Given a ψ -function, define a class of M-estimates via

$$\psi_c(r) = \psi(r/c)$$

- Tukey's Bi-weight.

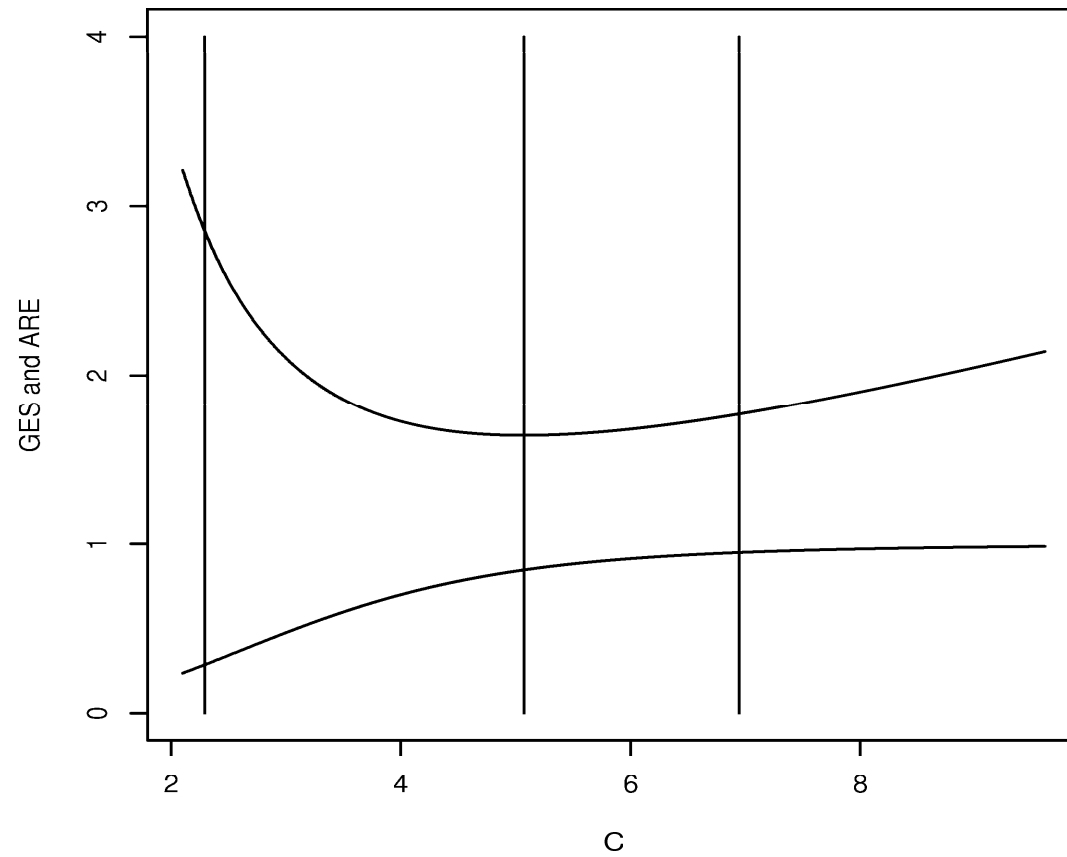
$$\psi(r) = r\{(1 - r^2)_+\}^2 \Rightarrow \psi_c(r) = \frac{r}{c} \left\{ \left(1 - \frac{r^2}{c^2}\right)_+ \right\}^2$$



- $c \rightarrow \infty \Rightarrow \approx$ Mean
- $c \rightarrow 0 \Rightarrow$ Locally very unstable

Normal distribution. Auxiliary scale.

GES and ARE for Bi-Weight M-estimates of Location



MORE THAN ONE OUTLIER

- GES: measure of local resistance/robustness.
- Measures of global resistance/robustness?

$\mathcal{X}_n = \{X_1, \dots, X_n\}$: n “good” data points

$\mathcal{Y}_m = \{Y_1, \dots, Y_m\}$: m “bad” data points

$\mathcal{Z}_{n+m} = \mathcal{X}_n \cup \mathcal{Y}_m$: ϵ_m -contaminated sample. $\epsilon_m = \frac{m}{n+m}$

Bias of a statistic: $|T(\mathcal{X}_n \cup \mathcal{Y}_m) - T(\mathcal{X}_n)|$

MAX-BIAS and THE BREAKDOWN POINT

- Max-Bias under ϵ_m -contamination:

$$B(\epsilon_m; T, \mathbf{X}_n) = \sup_{\mathcal{Y}_m} | T(\mathcal{X}_n \cup \mathcal{Y}_m) - T(\mathcal{X}_n) |$$

- Finite sample contamination breakdown point: *Donoho and Huber (1983)*

$$\epsilon_c^*(T; \mathbf{X}_n) = \inf\{\epsilon_m \mid B(\epsilon_m; T, \mathbf{X}_n) = \infty\}$$

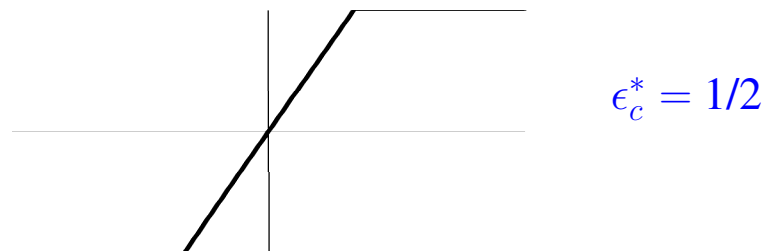
- Other concepts of breakdown (e.g. replacement).
- The definition of BIAS can be modified so that $\text{BIAS} \rightarrow \infty$ as $T(\mathcal{X}_n \cup \mathcal{Y}_m)$ goes to the boundary of the parameter space.

Example: If T represents scale, then choose

$$\text{BIAS} = | \log\{T(\mathcal{X}_n \cup \mathcal{Y}_m)\} - \log\{T(\mathcal{X}_n)\} |.$$

EXAMPLES

- Mean: $\epsilon_c^* = \frac{1}{n+1}$
- Median: $\epsilon_c^* = 1/2$
- α -trimmed mean: $\epsilon_c^* = \alpha$
- α -Windsorized mean: $\epsilon_c^* = \alpha$
- M-estimate of location with monotone and bounded ψ function:



PROOF

(sketch of lower bound)

Let $K = \sup_r |\psi(r)|$.

$$\begin{aligned} 0 &= \sum_{i=1}^{n+m} \psi(z_i - T_{n+m}) = \sum_{i=1}^n \psi(x_i - T_{n+m}) + \sum_{i=1}^m \psi(y_i - T_{n+m}) \\ &\quad \left| \sum_{i=1}^n \psi(x_i - T_{n+m}) \right| = \left| \sum_{i=1}^m \psi(y_i - T_{n+m}) \right| \leq mK \end{aligned}$$

Breakdown occurs $\Rightarrow |T_{n+m}| \rightarrow \infty$, say $T_{n+m} \rightarrow -\infty$

$$\Rightarrow \left| \sum_{i=1}^n \psi(x_i - T_{n+m}) \right| \rightarrow \left| \sum_{i=1}^n \psi(\infty) \right| = nK$$

Therefore, $m \geq n$ and so $\epsilon_c^* \geq 1/2$. $\square$

Population Version of Breakdown Point

under contamination neighborhoods

- Model Distribution: F • Contaminating Distribution: H
- ϵ -contaminated Distribution: $F_\epsilon = (1 - \epsilon)F + \epsilon H$
- Max-Bias under ϵ -contamination:

$$B(\epsilon; T, F) = \sup_H |T(F_\epsilon) - T(F)|$$

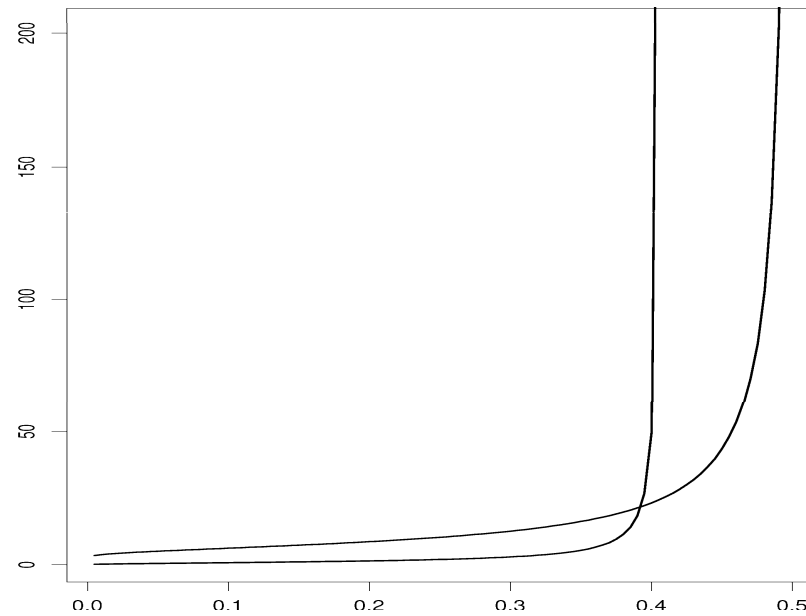
- Breakdown Point: *Hampel* (1968)

$$\epsilon^*(T; F) = \inf\{\epsilon \mid B(\epsilon; T, F) = \infty\}$$

- Examples

- Mean: $T(F) = E_F(X) \Rightarrow \epsilon^*(T; F) = 0$
- Median: $T(F) = F^{-1}(1/2) \Rightarrow \epsilon^*(T; F) = 1/2$

ILLUSTRATION



$$GES \approx \partial B(\epsilon; T, F) / \partial \epsilon \big|_{\epsilon=0}$$
$$\epsilon^*(T; F) = asymptote$$

Heuristic Interpretation of Breakdown Point

subject to debate

- Proportion of bad data a statistic can tolerate before becoming arbitrary or meaningless.
- If 1/2 the data is bad then one cannot distinguish between the good data and the bad data? $\Rightarrow \epsilon \leq 1/2?$
- Discussion: Davies and Gather (2005), *Annals of Statistics*.

Example: Redescending M-estimates of location with fixed scale

$$T(n_1, \dots, x_n) = \arg \min_t \sum_{i=1}^n \rho \left(\frac{|x_i - t|}{c} \right)$$

Breakdown point. *Huber (1984).* For bounded increasing ρ ,

$$\epsilon_c^*(T; F) = \frac{1 - A(\mathcal{X}_n; c)/n}{2 - A(\mathcal{X}_n; c)/n}$$

where $A(\mathcal{X}_n; c) = \min_t \sum_{i=1}^n \rho\left(\frac{x_i - t}{c}\right)$ and $\sup \rho(r) = 1$.

- Breakdown point depends on $\mathcal{X}_n$ and c
- $\epsilon^* : 0 \rightarrow 1/2$ as $c : 0 \rightarrow \infty$
- For large c , $T(n_1, \dots, x_n) \approx \text{Mean}!!$

Explanation: Relationship between redescending M-estimates of location and kernel density estimates

Chu, Glad, Godtliebsen and Marron (1998)

- **Objective function:**

$$\sum_{i=1}^n \rho \left(\frac{x_i - t}{c} \right)$$

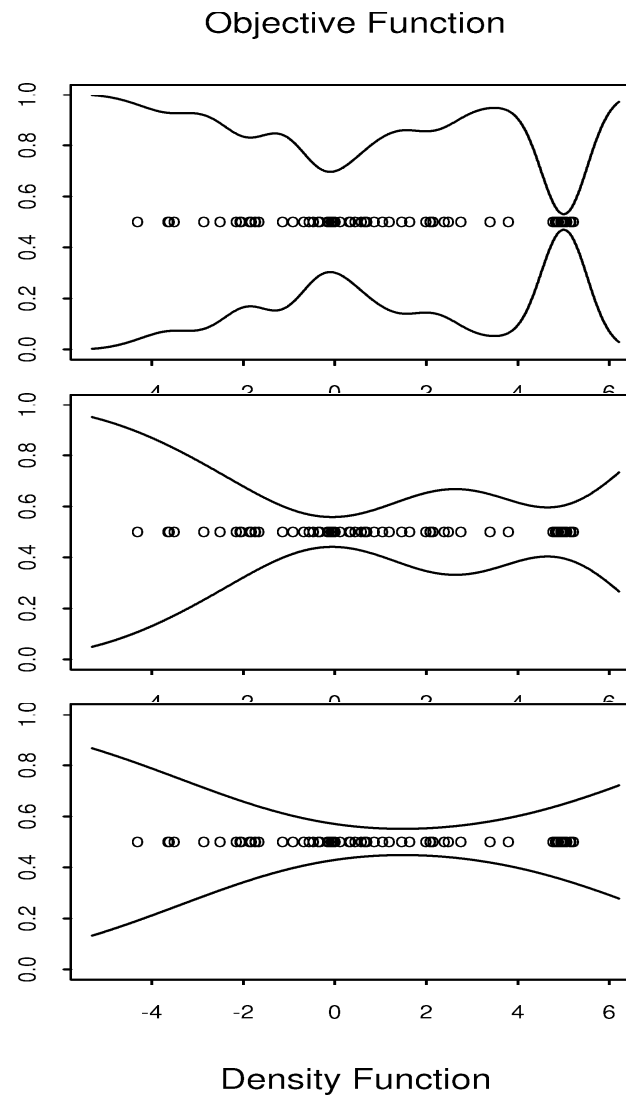
- **Kernel density estimate:**

$$\widehat{f}(x) \propto \sum_{i=1}^n \kappa \left(\frac{x_i - t}{h} \right)$$

- **Relationship:** $\kappa \propto 1 - \rho$ and $c = \text{window width } h$

- **Example:** $\kappa(r) = \frac{1}{\sqrt{2\pi}} \exp^{-r^2/2} \Rightarrow \rho(r) = 1 - \exp^{-r^2/2}$
Normal kernel $\Rightarrow$ Welsch's M-estimate

- **Example:** Epanechnikov kernel $\Rightarrow$ skipped mean



- “Outliers” less compact than “good” data $\Rightarrow$ Breakdown will not occur.
- Not true for monotonic M-estimates.

PART 3
ROBUST REGRESSION

REGRESSION SETTING

- Data: $(Y_i, \mathbf{X}_i)$ $i = 1, \dots, n$
 - $Y_i \in \mathfrak{R}$ Response
 - $\mathbf{X}_i \in \mathfrak{R}^p$ Predictors
- Predict: Y by $\mathbf{X}'\boldsymbol{\beta} = \sum_{j=1}^p x_j \beta_j$
- Residual for a given $\boldsymbol{\beta}$: $r_i(\boldsymbol{\beta}) = y_i - \mathbf{x}_i' \boldsymbol{\beta}$
- **HW:** Find MLE for $\boldsymbol{\beta}$ and σ^2 under model $Y_i = \mathbf{X}_i' \boldsymbol{\beta} + \epsilon_i$ where ϵ_i are i.i.d. $N(0, \sigma^2)$ using the theory of exponential families.
- M-estimates of regression
 - Generalization of MLE for symmetric error term
 - $Y_i = \mathbf{X}_i' \boldsymbol{\beta} + \epsilon_i$ where ϵ_i are i.i.d. symmetric.

M-ESTIMATES OF REGRESSION

- **Objective function approach:**

$$\min_{\boldsymbol{\beta}} \sum_{i=1}^n \rho(r_i(\boldsymbol{\beta}))$$

where $\rho(r) = \rho(-r)$ and $\rho \uparrow$ for $r \geq 0$.

- **M-estimating equation approach:**

$$\sum_{i=1}^n \psi(r_i(\boldsymbol{\beta})) \mathbf{x}_i = 0$$

e.g. $\psi(r) = \rho'(r)$

- **Interpretation: Adaptively weighted least squares.**

- Express $\psi(r) = ru(r)$ and $w_i = u(r_i(\widehat{\boldsymbol{\beta}}))$

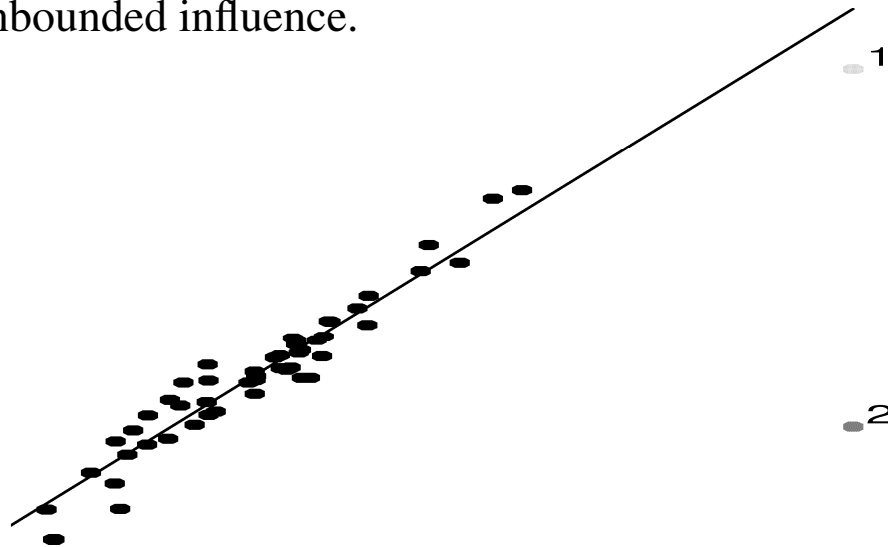
$$\widehat{\boldsymbol{\beta}} = \left[\sum_{i=1}^n w_i \mathbf{x}_i \mathbf{x}_i' \right]^{-1} \left[\sum_{i=1}^n w_i y_i \mathbf{x}_i \right]$$

- **Computations via IRLS algorithm.**

INFLUENCE FUNCTIONS FOR M-ESTIMATES OF REGRESSION

$$IF(y, \mathbf{x}; T, F) \propto \psi(r) \mathbf{x}$$

- $r = y - \mathbf{x}'T(F)$
- Due to residual: $\psi(r)$ • Due to Design: $\mathbf{x}$
- $GES = \infty$, i.e. unbounded influence.



- **Outlier 1:** highly influential for any ψ function.
- **Outlier 2:** highly influential for monotonic but not redescending ψ .

BOUNDED INFLUENCE REGRESSION

- **GM-estimates** (*Generalized M-estimates*).

- Mallows-type (*Mallows, 1975*):

$$\sum_{i=1}^n w(\mathbf{x}_i) \psi(r_i(\boldsymbol{\beta})) \mathbf{x}_i = 0$$

Downweights outlying design points (leverage points), even if they are good leverage points.

- General form (*c.g. Maronna and Yohai, 1981*):

$$\sum_{i=1}^n \psi(\mathbf{x}_i, r_i(\boldsymbol{\beta})) \mathbf{x}_i = 0$$

- **Breakdown points.**

- M-estimates: $\epsilon^* = 0$
- GM-estimates: $\epsilon^* \leq 1/(p+1)$

LATE 70's - EARLY 80's

- **Open problem:** Is high breakdown point regression possible?
- **Yes.** Repeated Median. *Siegel (1982)*.
 - Not regression equivariate

- **Regression equivariance:** For $a \in \Re$ and A nonsingular

$$(Y_i, \mathbf{X}_i) \rightarrow (aY_i, A'\mathbf{X}_i) \Rightarrow \widehat{\boldsymbol{\beta}} \rightarrow aA^{-1}\widehat{\boldsymbol{\beta}}$$

i.e.

$$\widehat{Y}_i \rightarrow a\widehat{Y}_i$$

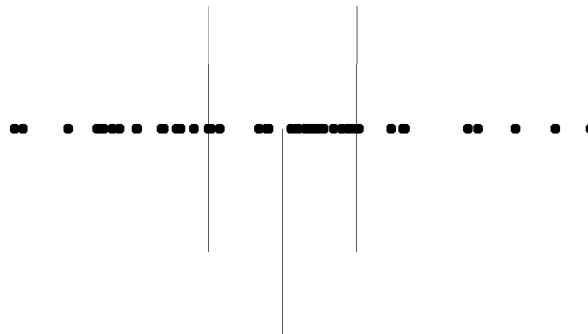
- **Open problem:** Is high breakdown point equivariate regression possible?
- **Yes.** Least Median of Squares. *Hampel (1984), Rousseeuw, (1984)*

Least Median of Squares (LMS)

Hampel (1984), Rousseeuw, (1984)

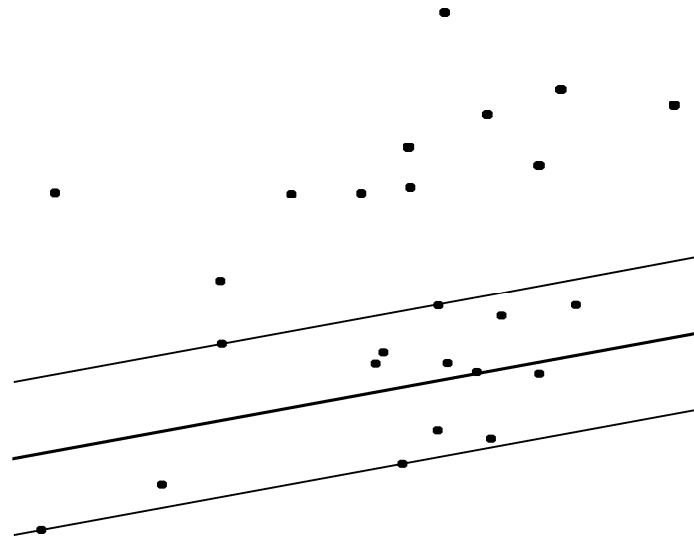
$$\min_{\beta} \text{Median}\{r_i(\beta)^2 \mid i = 1, \dots, n\}$$

- Alternatively: $\min_{\beta} \text{MAD}\{r_i(\beta)\}$
- Breakdown point: $\epsilon^* = 1/2$
- Location version: SHORTH (*Princeton Robustness Study*)



Midpoint of the **SHORTest Half**.

LMS: Mid-line of the shortest strip containing 1/2 of the data.



- **Problem:** Not $\sqrt{n}$ -consistent, but only $\sqrt[3]{n}$ -consistent

$$\sqrt{n} \| \hat{\beta} - \beta \| \rightarrow_p \infty$$

$$\sqrt[3]{n} \| \hat{\beta} - \beta \| = O_p(1)$$

- **Not locally stable.** e.g. *Example is pure noise.*

S-ESTIMATES OF REGRESSION

Rousseeuw and Yohai (1984)

- For $S(\cdot)$, an estimate of scale (about zero): $\min_{\beta} S(r_i(\beta))$
- $S = MAD \Rightarrow$ LMS
- $S =$ sample standard deviation about 0 $\Rightarrow$ Least Squares
- Bounded monotonic M-estimates of scale (about zero):

$$\sum_{i=1}^n \chi(|r_i|/s) = 0$$

for $\chi \uparrow$, and bounded above and below. Alternatively,

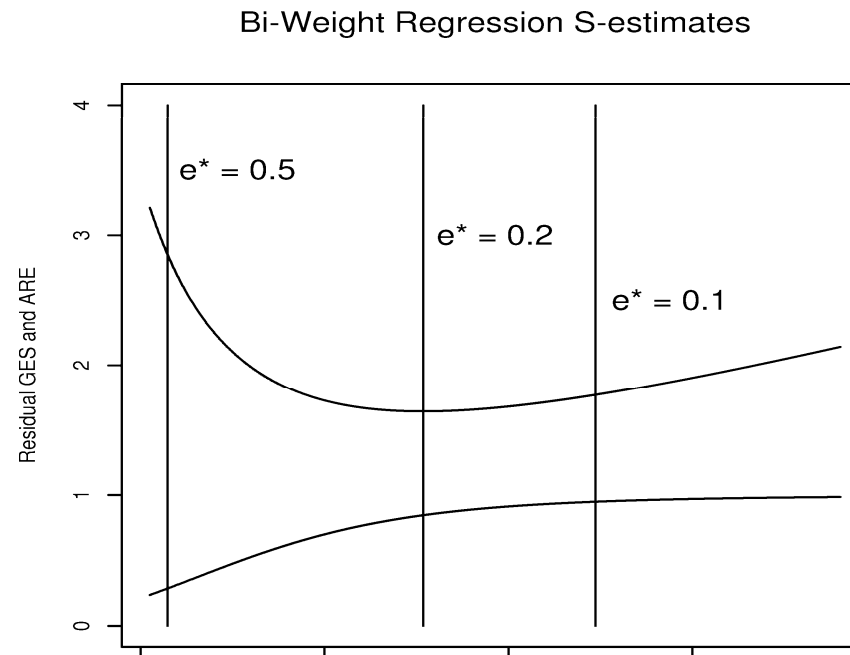
$$\frac{1}{n} \sum_{i=1}^n \rho(|r_i|/s) = \epsilon$$

for $\rho \uparrow$, and $0 \leq \rho \leq 1$

- For LMS: $\rho : 0 - 1$ jump function and $\epsilon = 1/2$
- Breakdown point: $\epsilon^* = \min(\epsilon, 1 - \epsilon)$

S-estimates of Regression

- $\sqrt{n}$ -consistent and Asymptotically normal.
- **Trade-off:** Higher breakdown point $\Rightarrow$ Lower efficiency **and** higher residual gross error sensitivity for Normal errors.



- **One Resolution:** One-step M-estimate via IRLS. (Note: R, S-plus, Matlab.)

M-ESTIMATES OF REGRESSION WITH GENERAL SCALE

Martin, Yohai, and Zamar (1989)

$$\min_{\beta} \sum_{i=1}^n \rho \left(\frac{|y_i - \mathbf{x}_i' \beta|}{c s_n} \right)$$

- Parameter of interest: β
- Scale statistic: s_n (consistent for σ at normal errors)
- Tuning constant: c
- Monotonic bounded $\rho \Rightarrow$ redescending M-estimate
- **High Breakdown Point Examples**
 - LMS and S-estimates
 - MM-estimates, Yohai (1987).

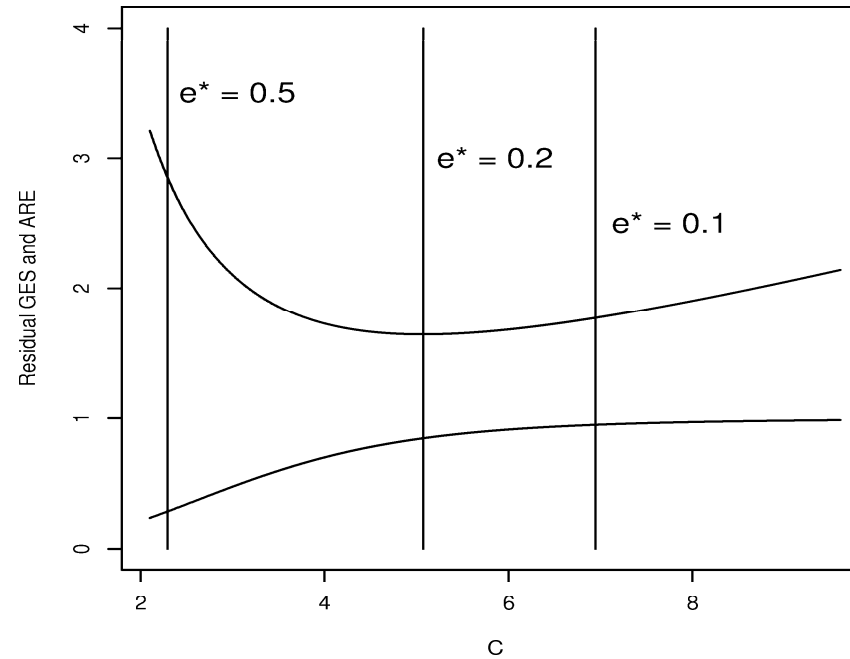
MM-ESTIMATES OF REGRESSION

Default robust regression estimate in R, S-plus

- Given a preliminary estimate of regression with breakdown point $1/2$.
Usually an S-estimate of regression.
- Compute a monotone M-estimate of scale about zero for the residuals.
 s_n : *Usually from the S-estimate.*
- Compute an M-estimate of regression using the scale statistics s_n and with any desired tuning constant c .
- Tune to obtain reasonable ARE and residual GES.
- Breakdown point: $\epsilon^* = 1/2$

TUNING

Bi-Weight Regression M-estimates



- High breakdown point S-estimates are badly tuned M-estimates.
- Tuning the S-estimates affects its breakdown point.
- MM-estimates can be tuned without affecting the breakdown point

COMPUTATIONAL ISSUES

- All known high breakdown point regression estimates are computationally intensive. *Non-convex optimization problem.*
- Approximate or stochastic algorithms.
- Random Elemental Subset Selection. *Rousseeuw* (1984).
 - Consider exact fit of lines for p of the data points $\Rightarrow \binom{n}{p}$ such lines.
 - Optimize the criterion over such lines.
 - For large n and p , randomly sample from such lines so that there is a good chance, say e.g. 95% chance, that one of the elemental subsets will contain only good data even if half the data is contaminated.